

Caelum Kaelin (617) 555-1212 | caelum.kaelin@email.com | linkedin.com/in/caelumkaelin | github.com/caelumkaelin

Professional Summary

Highly motivated and results-oriented Machine Learning Research Fellow at Aetherium Analytics with a proven track record in developing and deploying innovative AI/ML solutions. Possesses a strong academic background from MIT and expertise in various programming languages and machine learning frameworks, including Go, C++, R, TensorFlow, Keras, and Microsoft Azure. Passionate about leveraging cutting-edge technologies to solve complex problems and contribute to advancements in the field of artificial intelligence. Seeking a challenging role where I can apply my skills and experience to contribute to a dynamic and innovative team.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA * Master of Science in Computer Science, Artificial Intelligence Specialization (May 2021) * GPA: 3.9/4.0 * Thesis: "Novel Applications of Deep Reinforcement Learning in Personalized Medicine" * Bachelor of Science in Computer Science (May 2019) * GPA: 3.8/4.0 * Dean's List: All Semesters

Technical Skills

- **Programming Languages:** Go, C++, R, Python (NumPy, Pandas, Scikit-learn)
- **Machine Learning Frameworks:** TensorFlow, Keras, PyTorch
- **Cloud Platforms:** Microsoft Azure (Azure Machine Learning, Azure Databricks), AWS (Amazon SageMaker)
- **Databases:** SQL, NoSQL (MongoDB)
- **Big Data Technologies:** Spark, Hadoop
- **Version Control:** Git
- **Operating Systems:** Windows, macOS, Linux

Professional Experience

Machine Learning Research Fellow | Aetherium Analytics | Boston, MA | June 2021 – Present

- Developed and implemented novel deep learning models for anomaly detection in financial transactions, resulting in a 15% reduction in fraudulent activities.
- Designed and deployed a real-time recommendation system using collaborative filtering and deep reinforcement learning, increasing user engagement by 20%.

- Conducted extensive research on explainable AI (XAI) techniques and applied them to improve the transparency and interpretability of machine learning models.
- Contributed to the development of a scalable and robust machine learning pipeline using Azure Machine Learning.
- Mentored junior researchers and provided technical guidance on various machine learning projects.
- Presented research findings at internal and external conferences.

Software Engineering Intern | NovaTech Solutions | Cambridge, MA | June 2020 – August 2020

- Developed and maintained several key features for a high-traffic web application using Go and React.
- Implemented unit and integration tests to ensure the quality and reliability of the software.
- Collaborated with a team of engineers to design and implement a new microservice architecture.

AI/ML Projects

- **Personalized Medicine Recommendation System:** Developed a deep reinforcement learning agent that recommends personalized treatment plans for patients based on their medical history and genetic information. Utilized TensorFlow and Keras. (GitHub Repository: github.com/caelumkaelin/personalized-medicine)
- **Anomaly Detection in Financial Transactions:** Implemented a convolutional autoencoder model to detect fraudulent transactions in a large dataset of financial transactions. Achieved 95% accuracy in identifying fraudulent activities. Utilized TensorFlow and Azure Machine Learning. (GitHub Repository: github.com/caelumkaelin/fraud-detection)
- **Image Classification using Transfer Learning:** Developed a highly accurate image classification model using transfer learning with pre-trained ResNet models. Achieved state-of-the-art results on a challenging image classification dataset. Utilized PyTorch and AWS SageMaker. (GitHub Repository: github.com/caelumkaelin/image-classification)